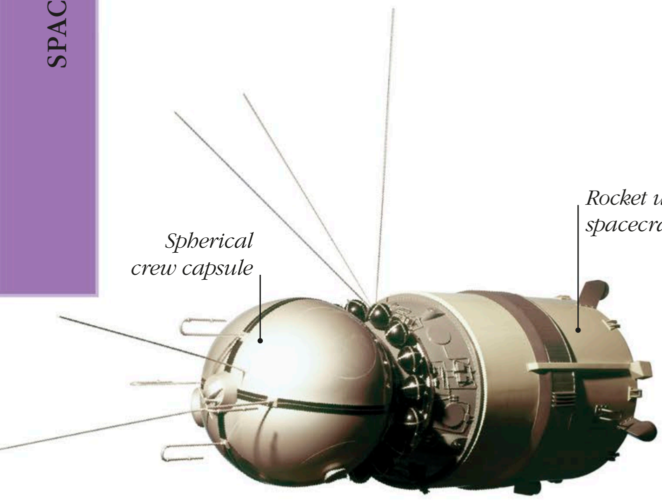
Manned spaceflight

Putting humans into space and bringing them back alive is the ultimate spaceflight challenge. Crewed spacecraft are heavier and more complex than most satellites, since they must carry equipment to keep astronauts alive for the duration of their missions, and protect them during the potentially hazardous return to Earth.

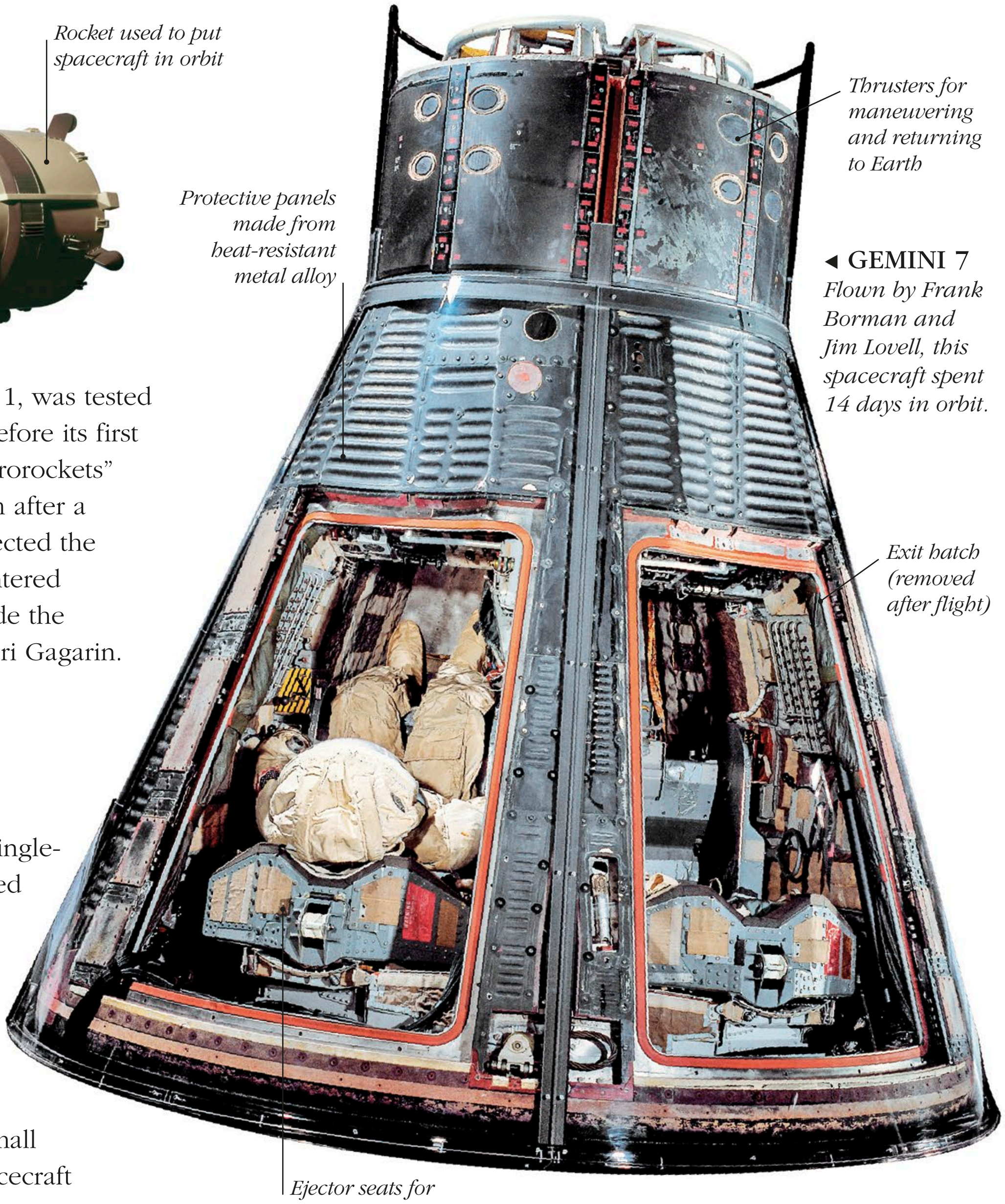
WOW!

The crews of Gemini 6A and Gemini 7 steered their spacecraft to within 12 in (30 cm) of each other in orbit.



VOSTOK 1

The first crewed spacecraft, Vostok 1, was tested with several unmanned launches before its first manned flight. Backward-firing "retrorockets" were used to return Vostok to Earth after a single orbit, and a heat shield protected the capsule from burning up as it re-entered the atmosphere at high speed. Inside the capsule was Russian cosmonaut Yuri Gagarin.



◂ **GEMINI 7**
Flown by Frank Borman and Jim Lovell, this spacecraft spent 14 days in orbit.

GEMINI

The first manned spacecraft were single-person vehicles, and mostly followed short, preset flight plans. NASA's Gemini spacecraft of the mid-1960s was a major advance on both fronts: it could carry two astronauts for missions of up to two weeks, and it could adjust its orbit with a system of thrusters—small rockets capable of pushing the spacecraft in various directions.